A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 13

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Other MEDCs (USA, Canada, Australia) also had surpluses – Japan already had suppliers **1** Eastern Europe / USSR needed food but could not pay **1** Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** Quotas limiting production (eg milk) **1** Set-aside payments for taking land out of production **1** Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) **1** Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
01.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape **1** Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat **1** Beetle banks of tussocky grass - natural pest predators **1** Skylark nesting plots – wild bird seed / nectar plants - winter food and pollinators **1** In-field tree protection – controlled stocking density to reduce compaction – ditch management – dry-stone walls – woodland management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) **1** The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino **1** Very low success rates with wildlife so far – deep-frozen cells could allow clones even after the last individuals die **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 3 points below. **1 Micro-propagation:** tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant **1 Black poplar:** cuttings / pollarded stems of pure native trees are rooted to produce many identical trees **1** Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
03.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild **1** Seeds collected worldwide – target of 10 000 seeds per species to maintain a diverse gene pool **1** Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident **1** Seeds germinated periodically to check viability and replenish stocks – can be used for reintroduction / breeding **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	
04.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Anchovy r-selected – blue whale k-selected **1** r-selected: mature quickly, produce many young, disperse widely, short lives – adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) **1** k-selected: mature late, few young with parental care, long-lived – populations stable near carrying capacity (whales, elephants, rhinos) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly **1** An increase in death rate above the low birth rate causes a population crash **1** Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium) – r-selected species rebound within a few years once exploitation stops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
05.3	1 mark for the point below. **1** New population = starting population + births + immigrants - deaths - emigrants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.4	**1 mark each** 1 mark for each of the 3 points below. **1** The greatest rate of exploitation (harvest) possible without causing long-term population decline **1 Equals the surplus production:** highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) **1** At carrying capacity net growth is zero so any harvest reduces the population – harvesting above MSY causes decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock • **sustainable** → meeting present needs without stopping future generations meeting theirs • **carrying capacity** → the largest population the area can support **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	
06.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2** Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance – indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met **up to 2 Single species:** easy planting, thinning and felling over large areas – but each tree species supports its own community so few species overall **up to 2** Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them **up to 2 Simple age structure:** uniform size makes management easy but gives little variety of light, temperature, wind – no old trees or dead wood for invertebrates and woodpeckers **up to 2 Mixed woodland:** varied niches, stratification, dead wood, indigenous trees each with their own dependent species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	6

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline **1** Continuing to grow them ties up land and capital for little extra timber – natural forests keep old and dead trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp **1** Wood fuel replaced by fossil fuels / electricity – so traditional management declined and woods became uniform **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors – many rare species with small ranges **1** High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) **1** Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches – intense competition for light drives specialisation **1** Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships – plants flower and fruit at different times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	4
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Adapted to stable conditions so they have narrow ranges of tolerance – a small change can move conditions outside the range **1** Many species are rare with small ranges / small populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) **1** Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Over half the Amazon's rainfall comes from transpiration within the forest – clearance reduces downwind rainfall **1** Rapid decay recycles nutrients so soils are shallow / poor – canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its mass	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced **1** The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze **1** Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Subsistence maize:** 9.0 / 0.6 = 15 **1 Intensive wheat:** 54.0 / 18.0 = 3 **1 Feedlot beef:** 12.0 / 40.0 = 0.3 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** An energy subsidy is the energy put into a farming system by people rather than by the Sun: fuel for machinery, the manufacture of fertilisers and pesticides, irrigation, transport, drying and storage **1** Intensive systems raise the yield per hectare by adding large energy subsidies, so the output rises but the input rises faster and the ratio falls **1** The feedlot ratio is below 1, so it takes more energy to produce the beef than the beef contains – the extra trophic level means most of the energy in the feed is lost in respiration, excretion and the inedible parts of the animal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Keep the food chain short by producing crops for direct human consumption rather than feeding them to livestock, so one transfer of about 10% efficiency is avoided **1** Restrict the movement of livestock, so less of the energy in the feed is used for muscular activity and more is converted to biomass **1** Keep the animals warm, or rear them indoors, so less energy is spent maintaining body temperature **1** Slaughter the animals young, while their growth rate and food conversion ratio are still high **1** Use species or breeds with a high food conversion ratio, such as poultry or farmed fish, rather than cattle **1** Control parasites and disease, and use feed that is easy to digest, so more of the food eaten is assimilated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** A single desired gene can be transferred, so the characteristic appears without the unwanted genes that come with a whole parent in conventional breeding **1** Genes can be moved between species that could never interbreed, so characteristics can be introduced that selective breeding could never produce, and the result is achieved in one generation instead of many **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
09.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Genes may escape by pollen transfer to wild relatives or to neighbouring non-GM crops, creating herbicide-resistant weeds or making organic certification impossible **1** Insect-resistant crops may kill non-target species, including the pollinators and natural pest predators that the wider ecosystem depends on **1** Pests and weeds evolve resistance to the toxin or the herbicide the crop is designed to be used with, so the advantage is temporary and use of the chemical increases **1** Reliance on a few patented GM varieties narrows the genetic diversity of the crop, making it more vulnerable to a new disease **1** Seed is patented and must be bought each year, which can put subsistence farmers in debt or make them dependent on one supplier **1** Long-term effects on human health are not fully known, and labelling may not let consumers choose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **genetic diversity** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	
10.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** By-catch = 5.9 + 1.9 = 7.8 t – total = 8.4 t **1** 7.8 / 8.4 x 100 = 92.9% (accept 93%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
10.2	1 mark for the point below. **1** The nets have a very small mesh size so few non-target animals can escape **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mesh size** → net hole size, set so that young fish escape	1

Do not write outside the box

Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 3 points below. **1** They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young **1** They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred **1** Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young – they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Impacts:** population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • **Catch quotas:** limit weight landed – work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • **Gear design:** mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag) – turtle exclusion devices – acoustic pingers for dolphins – curved hooks, decoys, sinkers and night setting to protect albatrosses – biodegradable / radio-tracked traps to reduce ghost fishing • **Restricted effort:** limits on boat size, engine power and days at sea – decommissioning – restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • **Evaluation:** enforcement at sea is difficult – some methods reduce catch or raise costs – quotas need reliable stock data – effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate – as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **quota** → a legal limit on the mass or number that may be caught	9
	• **mesh size** → net hole size, set so that young fish escape • **biodegradable** → broken down by decomposers **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Fishing's impacts go beyond depleting target species: k-selected species that breed late and produce few young collapse and recover slowly; by-catch of immature target fish, over-quota fish and non-commercial species is thrown back dead, with drift nets killing whales, dolphins and turtles, longlines drowning albatrosses and pair trawls porpoises; lost gear keeps ghost fishing; demersal trawls destroy sea fans, corals and seagrass; and removing any species alters its competitors, prey and predators.	
	Catch quotas limit the weight landed and work well for single-species pelagic shoals where fishing stops when the quota is reached, but in mixed demersal fisheries over-quota fish are discarded dead, so the quota does not protect them. Gear design tackles by-catch directly: mesh sizes and square-mesh escape panels let small fish out because, unlike diagonal mesh, they do not close under drag; turtle exclusion devices, acoustic pingers for dolphins, curved hooks, decoys, sinkers and night setting protect turtles, cetaceans and albatrosses; biodegradable and radio-tracked traps limit ghost fishing. Restricting effort - boat size, engine power, days at sea, decommissioning - reduces total pressure, and banning drift nets or demersal trawling on sensitive seabeds such as Rockall Bank protects habitats.	
	Effectiveness depends on enforcement at sea, which is difficult, on reliable stock data for quotas, on the species and fishery type, and on international cooperation because fish migrate; some measures cut catches or raise costs, and as local stocks fall fleets simply move further away. Gear modifications are the most effective per unit cost because they reduce damage without reducing legal catch; quotas and effort limits only work where they are enforced.	
	Weaker answer - Level 2 (5 marks)	
	Fishing damages the environment by overfishing, catching by-catch such as dolphins and damaging the seabed with trawling. Quotas limit how many fish can be caught so stocks can recover. Bigger mesh sizes let small fish escape so they can grow and breed. Turtle exclusion devices let turtles escape from nets. Some fishing methods can be banned in sensitive areas. No-take zones let fish breed. These methods help but they are hard to enforce at sea and some fishermen ignore them.	
	What makes the difference	
	• Level 2 covers impacts and methods soundly but without the mechanisms (why square mesh works, why quotas fail in mixed fisheries).	
	• Level 3 evaluates each method on its own terms and then compares them; Level 2's evaluation is a single general sentence.	
	• Named examples (Rockall Bank, albatross measures, ghost-fishing controls) give Level 3 the range the top band expects.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements – baseline and evaluation – Red List categorisation and quotas rely on it • **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR – sampling design (random / systematic, size, number, timing, standardisation) • **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked	25
	• **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Definition of sustainability (Brundtland) – population x per capita impact – dynamic equilibria, feedback, tipping points, diversity • **Principles of natural systems:** renewable low-density energy, low temperatures, material cycles, non-toxic wastes – contrast with linear human systems • **Circular economy principles:** cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • **Measuring progress:** ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • **Barriers:** economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	

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